(883607) 2016 TA56: a candidate dark comet on a Jupiter-family orbit

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Abstract

Dark comets are near-Earth objects whose orbits require non-gravitational accelerations of cometary magnitude although no coma is detected. We compare the 611 transverse non-gravitational terms that JPL has fitted to asteroid orbits with the largest acceleration absorbed sunlight can produce and with DAMIT spin poles. All 14 drifts with known poles have the sign expected from the spin. Five accelerations exceed this radiation limit at $500\,\mathrm{kg\,m^{-3}}$ and albedo 0.5: four belong to published dark comets or an active asteroid, and the fifth to (883607) 2016 TA56, on a Jupiter-family orbit ($T_J = 2.69$) and not previously listed among dark comets. A fit of its 102 positions gives $A_2 = (-3.45 \pm 0.37) \times 10^{-12}\,\mathrm{au\,d^{-2}}$, 9σ weighting each night as one position and 13σ under a night-level covariance fitted to the residuals, if a 2007 precovery is the same object; without that night, $A_2 = (-2.2 \pm 4.3) \times 10^{-12}\,\mathrm{au\,d^{-2}}$. The 2007 images support the identification: among all positions that orbits fitted to the later apparitions allow, the precovery source is the only one, nothing lies at the gravity-only position to 3.9 mag below it, and no other moving source shares its motion. NEOWISE gives a diameter of 1.4 to 3.7 km and an albedo of 0.050 to 0.007, at which the acceleration is 15 to 39 times the radiation limit at $1000\,\mathrm{kg\,m^{-3}}$. A single position in 2030 would measure A_2 to $\pm 0.4 \times 10^{-12}\,\mathrm{au\,d^{-2}}$ independently of the precovery. The screen recovers, retrospectively, the weakly active comet (875163) 1998 SH2 (Farnocchia et al., 2026), whose acceleration relative to the radiation limit is a twentieth of that of 2016 TA56.

Keywords: Comets, Near-Earth objects, Asteroids, Celestial mechanics, Orbit determination

1. Introduction

Astrometry can reveal departures from a purely gravitational orbit. The thermal re-emission of absorbed sunlight, delayed by the body's rotation, produces the Yarkovsky drift, which JPL's orbit solutions model as a transverse acceleration A_2 scaling as r^{-2} [5], and outgassing from an active comet produces larger accelerations [11]. Some near-Earth objects show accelerations too large for radiation forces but no detectable coma. These dark comets [16] divide into an inner population of small bodies on near-circular orbits and an outer population of larger bodies on eccentric, Jupiter-family-like orbits [17], and their accelerations imply outgassing at rates too low for the associated dust to be detected; Seligman et al. [17] list fourteen such objects. Farnocchia et al. [6] found that the 1998–2025 astrometry of the near-Earth object (875163) 1998 SH2 shows perturbations consistent with outgassing, detected a weak tail of low surface brightness during its 2025 close approach that archival images had not revealed, and proposed that non-gravitational perturbations over long astrometric arcs can identify further comets among near-Earth objects classified as asteroids.

We present such a search: a screen of every transverse term in the JPL Small-Body Database for accelerations larger than absorbed sunlight can produce (the radiation limit), and for drifts of

the wrong sign for the body's known spin. Of the thirteen published dark comets with a transverse term, the screen recovers the six whose terms exceed the radiation limit, and it identifies one further object, 2016 TA56; under typical assumptions for density and albedo it also flags 1998 SH2, recovering retrospectively a weakly active comet from its non-gravitational acceleration alone. Such a search was proposed by Farnocchia et al. [6]; this work contributes the screen and the candidate it identifies.

Section 2 describes the data and methods. Section 3 presents the screen, an independent orbit determination for 2016 TA56, the tests of its 2007 precovery in the archival images, its size and albedo from NEOWISE, the minimum outgassing rate, the 2030 apparition, an overview of the tests, and the same analysis of the two further objects that the screen identifies, 1998 SH2 and 2006 VC. Section 4 relates the results to previous work, and Section 5 summarises the conclusions.

2. Data and methods

The screen. The Small-Body Database query API returns every asteroid with a transverse term A_2 ($\mathrm{au\,d^{-2}}$ at 1 au) and its uncertainty, with H, any measured diameter and the orbital elements; we queried it on 2026 October 16 for every asteroid whose orbit solution includes A_2, and the query and its raw result are available in the repository given under Data availability. Each term is compared with an upper bound on the transverse acceleration that sunlight can impart. A sphere of diameter D at 1 au intercepts the power $S\pi D^2/4$, with $S = 1361\,\mathrm{W\,m^{-2}}$. Absorbed photons arrive from the Sun and exert a purely radial force, so a transverse force can come only from light that leaves the body, reflected or re-emitted; its momentum flux is at most the intercepted power divided by c, a value reached only if all of it leaves in one direction along the orbit. Dividing by the mass $\rho\pi D^3/6$ of a body of uniform bulk density ρ gives

$$a_{\max} = \frac{S\pi D^2/4}{c\,\rho\pi D^3/6} = \frac{3}{2}\,\frac{S/c}{\rho D}, \tag{1}$$

which we call the radiation limit, with $S/c = 4.54 \times 10^{-6}\,\mathrm{N\,m^{-2}}$. An efficiency $\xi = |A_2|/a_{\max}$ greater than one therefore requires a force other than radiation. The limit is a ceiling: thermal recoil re-emits most of the absorbed energy in directions that cancel, and a ratio below one does not show that radiation can supply an acceleration. We evaluate ξ at the smallest diameter allowed by H (albedo 0.5, unless a diameter has been measured) with $\rho = 500\,\mathrm{kg\,m^{-3}}$, and again at $2000\,\mathrm{kg\,m^{-3}}$ and albedo 0.14. The diurnal component of thermal recoil drives a prograde rotator outward and a retrograde rotator inward, while the seasonal component, largest at 90° obliquity, drives any rotator inward [21]. A retrograde rotator therefore cannot drift outward whatever its thermal properties, whereas a prograde rotator near 90° obliquity can drift inward. Each drift significant at 3σ is compared with the obliquity of every pole solution in DAMIT [4], with the orbit normal from the osculating inclination and node.

Astrometry. The Minor Planet Center lists 102 optical positions of 2016 TA56, from a precovery on 2007 September 14 to 2026 June 10, in six apparitions: 12 positions on one night in 2007, 79 from its discovery by Pan-STARRS on 2016 October 10 to 2017 June, and 2, 4, 3 and 2 in 2020 November, 2022 August, 2025 October and 2026 June. Twenty-three of them are NEOWISE positions from twenty exposures, each with the spacecraft's geocentric position on the second line of the record.

Orbit fits. We integrate orbits with ASSIST [8], which is built on REBOUND [14] and its IAS15 integrator [15], using the DE440 planetary ephemeris [13], the sixteen most massive asteroids, the

oblateness of the Sun and the Earth, and general relativity. The transverse acceleration is $A_2\,g(r)$ with $g(r) = r^{-2}$, as in JPL's solution. A ground site's position is computed from its MPC parallax constants and transformed into the GCRS with astropy [1]; a NEOWISE position uses the reported spacecraft position. Each model position is corrected for light time but not for aberration, which affects the reference stars of the reduction equally. The fit adjusts the barycentric state at TDB 2457746.5 (2016 December 24) and, where included, A_2, by Gauss–Newton iteration with forward-difference partial derivatives. Each site is weighted by its rms residual over its retained positions where it has five or more, with a floor of $0.15''$, and by $0.5''$ otherwise; a position whose two-coordinate χ^2 exceeds 8 is excluded at each iteration and readmitted if it falls below that value. Positions from the same night share the reference catalogue, the observer's clock and the seeing, so their errors are correlated, and we therefore repeat each fit under two further weightings: that of Vereš et al. [20], used in JPL's solutions, which inflates the uncertainties of more than four positions from one station on one night by $\sqrt{N/4}$, and one that inflates every position of a night by $\sqrt{N}$ across all stations, so that each night carries the weight of a single position. We fit gravity only, A_2, the radial term A_1 together with A_2, and A_2 with each apparition omitted in turn, and compare each apparition with the gravity-only orbit fitted to the others, decomposing its residuals along and across the apparent motion. We also estimate the correlation within a night from the residuals, with a covariance in which each position's site variance is scaled by s^2 and any two positions of one night share a variance τ^2 in each coordinate, s and τ set by restricted maximum likelihood. For the precovery night, the positions that the other apparitions allow are those of the orbit fitted without that night with A_2 held at each value from -16 to $+12 \times 10^{-12}\,\mathrm{au\,d^{-2}}$, the state at its conditional mean given A_2 under that fit's covariance; with A_1 also free, the covariance of the predicted position shows how far a radial term can move it.

Precovery images. We remeasured all sixteen CFHT MegaCam exposures of the precovery night (300 and 420 s), independently of the published reports. Each CCD is registered to Gaia DR3 stars with proper motions propagated to 2007, by a quadratic map from detector pixels to the pixels of the archive's world coordinate system. Cosmic rays are replaced by the median of their neighbourhood, and sources are found in each CCD with a Gaussian filter for point sources and with a filter trailed by the object's motion during the exposure. A source above 5σ is classed as static when three other frames contain a source above 3σ within $1.2''$ of it; the remaining sources are linked into straight tracks that require detections in six or more frames within $0.6''$ of the track, an rms residual of at most $0.3''$, and a source within $0.6''$ of the track in at least 60 per cent of the frames whose CCDs cover it. Repeating the linking with the frame times permuted, which destroys real tracks, gives the rate of spurious tracks, and synthetic copies of the object, trailed and scaled to its flux in each band, injected along 300 random tracks within $10''$ per hour and 20° of its motion and from 0.5 mag brighter to 2 mag fainter, give the completeness. To test each position at which an orbit could place the object, we sample the trailed filter's S/N there in every frame, divide it by the median S/N of the detection in that band, and take the median over the frames: the result is the flux, in units of the detection's, of any source moving with the object through that position, and a static source that crosses one or two frames does not change it. A frame's measurement is left out within $4'' + 3''(18 - G)$ of a Gaia star brighter than $G = 18$, within $2.5''$ of a static source above S/N 20, and within $10''$ of the edge of its CCD, and a position requires measurements in at least eight frames.

Thermal fit. For each NEOWISE position the single-exposure source table gives the W1 ($3.4\,\mu$m) and W2 ($4.6\,\mu$m) profile-fit magnitudes [22, 10], and Horizons gives the distances from the

Sun and from WISE and the phase angle. We fit the fluxes for the diameter D and the beaming parameter η with the near-Earth asteroid thermal model [7], with emissivity 0.9 and no night-side emission, plus reflected sunlight from the H–G phase law [3] ($G = 0.15$), a 5778 K solar spectrum and an infrared albedo 1.6 times the visible [9]; the visible albedo follows from H. Each flux carries its photometric uncertainty plus a 10 per cent floor for the colour corrections and the calibration.

3. Results

3.1. The screen

Of the 611 asteroids with a transverse term in the database on 2026 October 16, 567 have it at 3σ or more, and 14 of these have DAMIT pole solutions; Table S1 of the supplementary material lists all 611 with their terms, diameters, efficiencies, pole solutions and spin-sign checks. All 14 drifts have the sign expected from the spin: no retrograde rotator drifts outward and no prograde rotator inward, and for the four objects with two pole solutions both solutions give the same sense of rotation. The eleven retrograde rotators, with obliquities of 118 to 179°, all drift inward, which tests the signs of their terms directly; the three prograde rotators, at 10 to 48°, all drift outward, and only (1580) Betulia, at 118°, lies within 30° of 90°. Under the typical assumptions the 567 significant terms have a median efficiency of 0.075, and 95 per cent of them lie below 0.17. Five accelerations exceed the radiation limit at 500 kg m^{-3} and albedo 0.5 (Table 1): those of the active asteroid 362P/(2008 GO98), of the dark comets 2012 UR158, 2005 UY6 and (523599) 2003 RM [16, 17], and of 2016 TA56. At 2000 kg m^{-3} and albedo 0.14, ten exceed it, including the dark comets 1998 FR11, 2001 ME1 and 2006 RH120 and two objects absent from the dark-comet lists, 1998 SH2 and 2006 VC, with efficiencies of 1.6 and 1.3; Section 3.8 analyses both, and 1998 SH2 is a comet [6]. Only 2016 TA56 exceeds the limit under the conservative assumptions without appearing in a published list of dark comets or active asteroids. Thirteen of the fourteen published dark comets have a transverse term in the database, and the screen flags six of them. The other seven are the inner dark comets of 4 to 21 m, whose transverse terms are 0.01 to 0.09 of the radiation limit for bodies of their size and whose published accelerations are based on the radial and normal terms (five of the seven have A_3 at 3.6 to 10.7σ) and on thermal models of the bodies, which a transverse screen cannot test. The screen is therefore sensitive to large dark comets of the outer population, while the inner population is identified through the radial terms. The screen evaluates terms that JPL has already fitted, and JPL fits A_2 only where the astrometry requires or can constrain it: the 611 include 609 of the 42,772 near-Earth asteroids in the database, and two other objects. It finds dark comets only among bodies whose transverse accelerations are already measured, it says nothing about the other near-Earth objects, and the number it finds is not an estimate of how common dark comets are.

3.2. The orbit of 2016 TA56

JPL's solution 23 (2026 June 17) uses 94 of the positions and gives $A_2 = (-3.19 \pm 0.36) \times 10^{-12}$ au d^{-2}, an 8.9σ detection, with $a = 2.69$ au, $e = 0.782$, $q = 0.588$ au, aphelion 4.80 au, $i = 32.3°$, a minimum orbit intersection distance with Jupiter of 0.185 au and the most recent perihelion on 2026 January 13. Our fits (Table 2) reproduce the term. With site weights, the fit with A_2 retains 97 of the 102 positions at an rms of 0.33″ and gives $A_2 = (-3.54 \pm 0.17) \times 10^{-12}$ au d^{-2}; this uncertainty treats the twelve precovery positions as independent. With the same-night deweighting used in JPL's fits, which inflates the eight positions from site T14 by $\sqrt{2}$,

Table 1: Transverse accelerations above the radiation limit at $500\,\mathrm{kg\,m^{-3}}$ and albedo 0.5 (or at a measured diameter). $D_{\min}$ is the diameter used, ξ the ratio of the acceleration to $a_{\max}$ at that diameter, and ξ_{typ} the same ratio at $2000\,\mathrm{kg\,m^{-3}}$ and albedo 0.14.

Object	A_2 ($10^{-12}\,\mathrm{au\,d^{-2}}$)	S/N	$D_{\min}$ (m)	ξ	ξ_{typ}	T_J, status
362P/(2008 GO98)	−1150	11.2	4921	8333	62990	2.93, active
2012 UR158	−7.20	119	629	6.67	26.7	3.00, dark comet
(883607) 2016 TA56	−3.19	8.9	445	2.08	15.8	2.69, this work
(452639) 2005 UY6	−0.54	11.3	2249	1.78	7.12	2.94, dark comet
(523599) 2003 RM	+3.29	110	223	1.08	8.15	2.95, dark comet

the result is $(-3.56 \pm 0.19) \times 10^{-12}$; with the whole night weighted as a single position, the limit in which the two reports share all their errors, it is $(-3.45 \pm 0.37) \times 10^{-12}$, a 9σ detection consistent with JPL's uncertainty. The value is the same under all three weightings, and we adopt the last. The deweighted fits have χ^2/dof well below one because inflating the uncertainties of a night's positions gives them the weight of one, so their χ^2 reflects the weighting rather than the size of the errors, and the site-weighted fit's 0.65 reflects the $0.15''$ floor on sites whose scatter is smaller. A fourth treatment estimates the correlation from the residuals: each position's site variance is scaled by s^2, and any two positions of the same night share a variance τ^2 in each coordinate. Restricted maximum likelihood over the 97 positions kept on 28 nights gives $s = 0.68$ and $\tau = 0.17''$ (95 per cent interval 0.11 to $0.23''$), and with this covariance $A_2 = (-3.85 \pm 0.30) \times 10^{-12}$, a 13σ term, or $(-3.88 \pm 0.37) \times 10^{-12}$ with τ at its upper bound. Weighting each night as one position is therefore the more conservative treatment. These uncertainties belong to fits that include the 2007 night; the dependence on that night, and the possibility that its source is a different object, are examined in Section 3.3. Omitting one night at a time, the one-position-per-night fit gives A_2 between -3.96 and -3.39×10^{-12} for every night after 2007, a jackknife uncertainty of 0.53×10^{-12} that corresponds to 6.5σ; resampling those nights with replacement, with the 2007 night always included, gives a 95 per cent interval of -4.9 to -3.2×10^{-12} over 200 samples. Clipping thresholds of 6, 12 and none, error floors of 0.10 and $0.25''$, and default uncertainties of 0.3 and $0.8''$ change the term by at most 0.14×10^{-12}, with significances between 7.4 and 10.6σ. A radial term fitted simultaneously is $A_1 = (+3.7 \pm 1.9) \times 10^{-10}\,\mathrm{au\,d^{-2}}$, 2σ from zero, with $A_2 = (-3.32 \pm 0.21) \times 10^{-12}$: the radial term, which the geometry of a single precovery night constrains poorly, does not absorb the transverse one. The gravity-only fit retains 83 positions, rejecting all twelve from 2007 and all three from 2025 October.

3.3. Dependence on the 2007 precovery

Omitting the 2007 precovery gives $A_2 = (-2.2 \pm 4.3) \times 10^{-12}\,\mathrm{au\,d^{-2}}$, of the same sign but no longer significant; omitting any other apparition except the discovery apparition leaves it above 6σ. Resampling all 28 nights with replacement gives a term beyond 3σ in 57 per cent of 200 samples, close to the 64 per cent of samples that include the 2007 night, so nearly every significant resample contains it. The detection is therefore conditional on the precovery: with that night the term is measured at 9 to 13σ, according to the treatment of same-night errors, and without it the later apparitions do not measure it. Under the night-level covariance, the twelve 2007 positions would have to share a further error of $0.68''$ in each coordinate to bring A_2 down to 5σ, and of $1.20''$ to bring it to 3σ. The twelve precovery positions come from two reports from Mauna Kea (sites 568 and T14) for 2007 September 14, with four and eight positions interleaved over 78 minutes and

Table 2: Orbit fits of 2016 TA56 to the 102 MPC positions. The first three fits differ in the weighting of positions from the same night (Section 2).

Fit	Positions	Kept	χ^2/dof	A_2 (10^{-12} au d^{-2})
with A_2, site weights	102	97	121.2/187	-3.54 ± 0.17
with A_2, nights deweighted by $\sqrt{N/4}$	102	97	105.8/187	-3.56 ± 0.19
with A_2, one position a night	102	101	58.2/195	-3.45 ± 0.37
with A_1 and A_2	102	97	117.3/186	-3.32 ± 0.21, $A_1 = +367 \pm 187$
gravity only	102	83	112.8/160	—
without 2007 September	90	85	118.2/163	-2.24 ± 4.34
without 2016–2017	23	21	2.5/35	-4.80 ± 3.20
without 2020 November	100	95	120.4/183	-3.55 ± 0.17
without 2022 August	98	93	120.8/179	-3.54 ± 0.18
without 2025 October	99	95	116.1/183	-3.79 ± 0.57
without 2026 June	100	95	120.8/183	-3.54 ± 0.17

consecutive frames of each report at least 7.0 (568) and 5.8 minutes (T14) apart, in the D1 field of the CFHT Legacy Survey, where the predicted motion was 34.5″ per hour toward position angle 210°. Both reports are re-measurements of the same CFHT frames against Gaia DR2, so they share the frame timing and the reference catalogue: they are two reductions of one night, not two independent nights. The gravity-only orbit fitted to the later apparitions predicts the positions for that night with uncertainties of 0.08″ along the apparent motion and 0.20″ across it, from its covariance; the precovery positions lie 3.45″ from that prediction, 1.8″ along the motion and 3.0″ across it, with a scatter of 0.07″ (Figure 1a).

A timing error displaces a position only along the apparent motion. It could therefore account for the along-track offset, which corresponds to three minutes, but not for the cross-track offset, which is fifteen times the prediction uncertainty whether or not the two reports share a clock. Both reports use Gaia DR2 as the reference catalogue, which rules out a catalogue offset of this size. We measured the CFHT frames independently (Table 3, Figure 2), registering each to Gaia DR3 stars with proper motions propagated to 2007, with a median rms of 0.02″. The object is detected in twelve frames, each time trailed by its motion during the exposure (4.06″ in the 420 s r frames and 2.90″ in the 300 s g frames), along a straight track from which both reports deviate by 0.10″ rms across the motion. In eleven frames our centroid lies within 0.42″ of the accelerated orbit and 3.2 to 3.4″ east of the gravity-only orbit; in frame 942303, where the object is faint, it lies 1.7″ away, and in frame 942308, where the object is next to a star, it is not detected. The offset is therefore present in the images rather than introduced by either reduction, and removing the transverse term would require a common error of 3.0″ across the motion in every position, 150 times the scatter of the registration. With A_2 the precovery positions are fitted to within 0.07″ (Figure 1c).

The later positions support the same conclusion. A gravity-only orbit fitted to the precovery and the apparitions through 2022 fits the precovery but misses the three positions of 2025 October by 10.5″ (6.7″ along and 8.0″ across the motion), against a prediction uncertainty of 0.4″, and the two of 2026 June by 2.3″, which the fit then rejects (Figure 1b). No gravity-only orbit fits the precovery and the last two apparitions together, while the orbit with the transverse term fits all of them. The term is therefore robust unless the precovery is a different object. A misidentification would leave a body whose 2016–2026 arc constrains A_2 only to $\pm 4.3 \times 10^{-12}$ au d^{-2}, and strong

non-gravitational accelerations can themselves complicate the linking of observations [19]. The three reductions of that night agree to 0.2″, and the measured motion, 34.7″ per hour toward position angle 209°, is the rate and direction predicted by the orbit.

The 2007 frames test the identification directly, because 2016 TA56 must appear in them whatever its acceleration. Every orbit fitted to the later apparitions places it on one line through the gravity-only position, 0.93″ further west for each 10^{-12} au d^{-2} of A_2, and a radial term fitted as well moves it only along the same line (1σ ellipse $4.05'' \times 0.17''$). Its absolute magnitude predicts $V = 23.1$ that night, and the measured source has $r = 22.8$. Along this line the frames show a single source (Figure 3): the median flux reaches 0.96 of the detection's at $A_2 = -3.75 \times 10^{-12}$ au d^{-2} and is nowhere else above 0.05, and within 24″ of the gravity-only position in any direction nothing but the detection exceeds 0.13. At the gravity-only position it is −0.04, while lines displaced across the field give a robust scatter of 0.022 at the same point, so any object there is at least 3.9 mag fainter than the detection (3σ). Of the 954 displaced lines, none reach half the detection's flux anywhere along their length. The precovery source is therefore 2016 TA56 unless that object was about 40 times fainter than its absolute magnitude implies while an unrelated object moving as predicted lay on the line.

A census of the moving sources in all sixteen frames measures the density of such objects in these images, rather than among known asteroids. It finds 68 moving sources besides 2016 TA56 over the 0.86 deg^2 of the mosaic, 79 per square degree, among them 21 of the 22 known asteroids with good positions that SkyBoT [2] places on the frames. They move at $22 \pm 13''$ per hour; none comes within 5″ per hour and 10° of the object's motion, and three within 10″ per hour and 20°. Injected copies of the object are recovered on 109 of the 115 tracks placed within 0.5 mag of its brightness, and no track forms with the frame times shuffled. With normal distributions fitted to the rates and directions of the census, the expected number of unrelated objects within 30″ of the predicted position and within 0.5″ per hour and 1° of the measured motion is 8×10^{-6}.

The 2007 night is the only epoch before the discovery that constrains the orbit. The other images of the predicted track listed by the CADC SSOIS index recorded the object at $V = 23.2$ or fainter, beyond the depth of their exposures: forced photometry at the predicted position gives S/N 0.9 and 2.2 in the deepest, two DECam exposures of 2013 September 29 at $V = 23.9$, and 1.4 in a stack of eight Pan-STARRS1 exposures of 2013 August, and the accelerated and gravity-only orbits are only 0.5″ apart in 2013.

3.4. Size, albedo and the radiation bound

NEOWISE detected 2016 TA56 twenty times: eight times in 2016 December at 1.61 to 1.62 au from the Sun and twelve times in 2017 June at 1.39 to 1.40 au, at phase angles of 37 to 47°. Sixteen W2 fluxes reach S/N 4 after the rejection of one that differs by a magnitude from the rest of its apparition. At $\eta = 1.4$ the fit gives $D = 2.44 \pm 0.06$ km and $p_V = 0.017$, with $\chi^2 = 28.4$ for 16 fluxes, an uncertainty conditional on that value of η; η from 0.9 to 1.9 gives $D = 1.41$ to 3.72 km and $p_V = 0.050$ to 0.007 with nearly equal χ^2, because W2 alone does not constrain η (Table 4), and this range is the uncertainty of the diameter. Of the other assumptions, an emissivity of 0.95 increases D by 5 per cent, while the phase slope ($G = 0.05$ or 0.25), the ratio of infrared to visible albedo (1.0 or 2.0) and the flux floor (5 or 20 per cent) change it by less than 1 per cent. The ten W1 measurements that reach S/N 4 lie 0.6 to 4.5 mag above the model and scatter by two magnitudes within an apparition, and one of them coincides with a W2 measurement at S/N 2, as expected for faint single-exposure detections and blends with stars; they are not used. Table S2 of the supplementary material lists all twenty detections with their W1 and W2

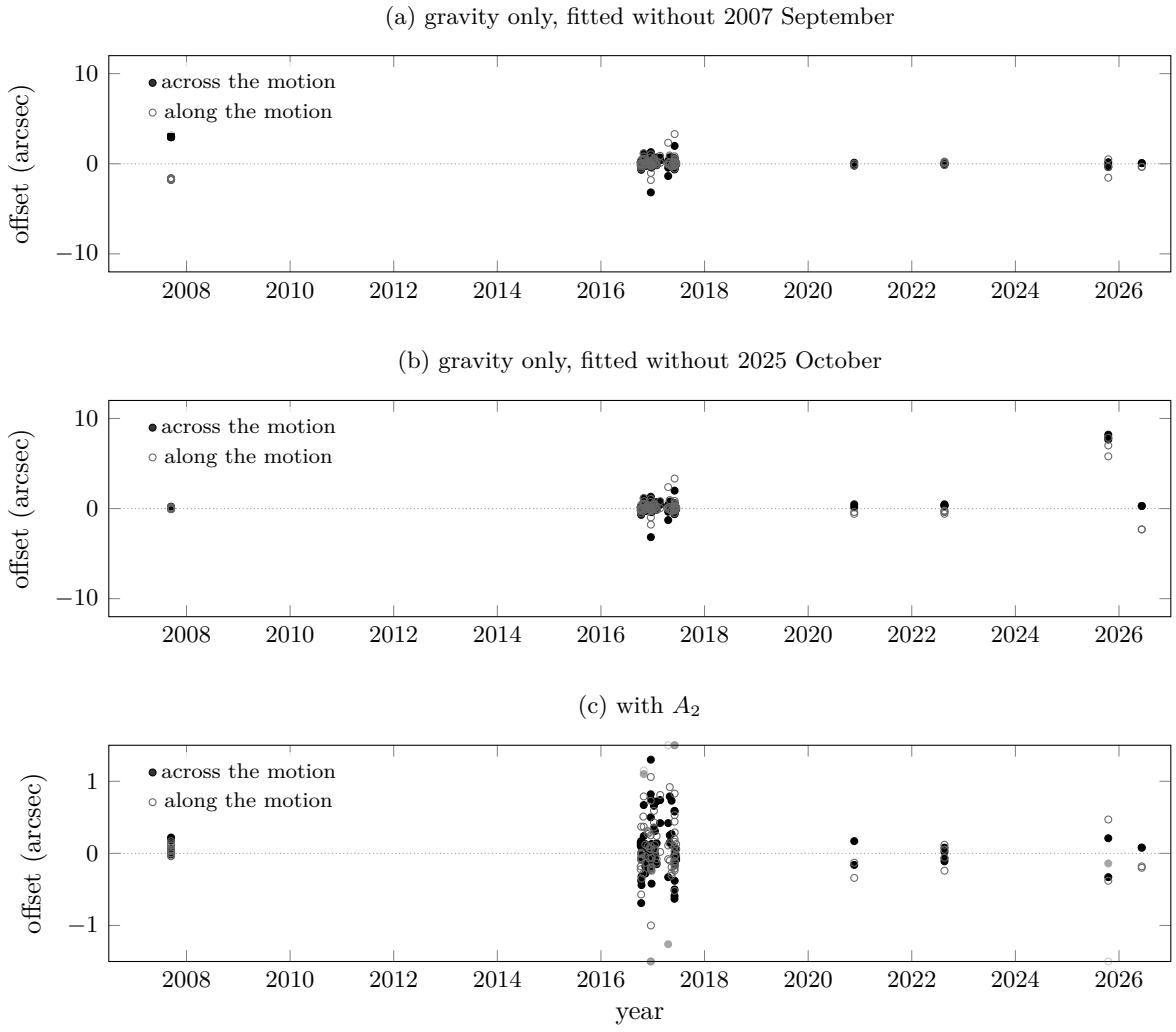


Figure 1: Residuals of each position from three orbits, across and along the apparent motion, as a function of time: (a) the gravity-only orbit fitted to the 2016–2026 apparitions, which is offset from the 2007 precovery by 3.0″ across the motion; (b) the gravity-only orbit fitted to the precovery and the apparitions through 2022 (the 2026 positions rejected), which is offset from the 2025 October positions by 8″ across and 7″ along the motion; (c) the orbit with A_2, with all residuals within 1.5″ and rejected positions shown faint. Positions from the same night overlap at this scale.

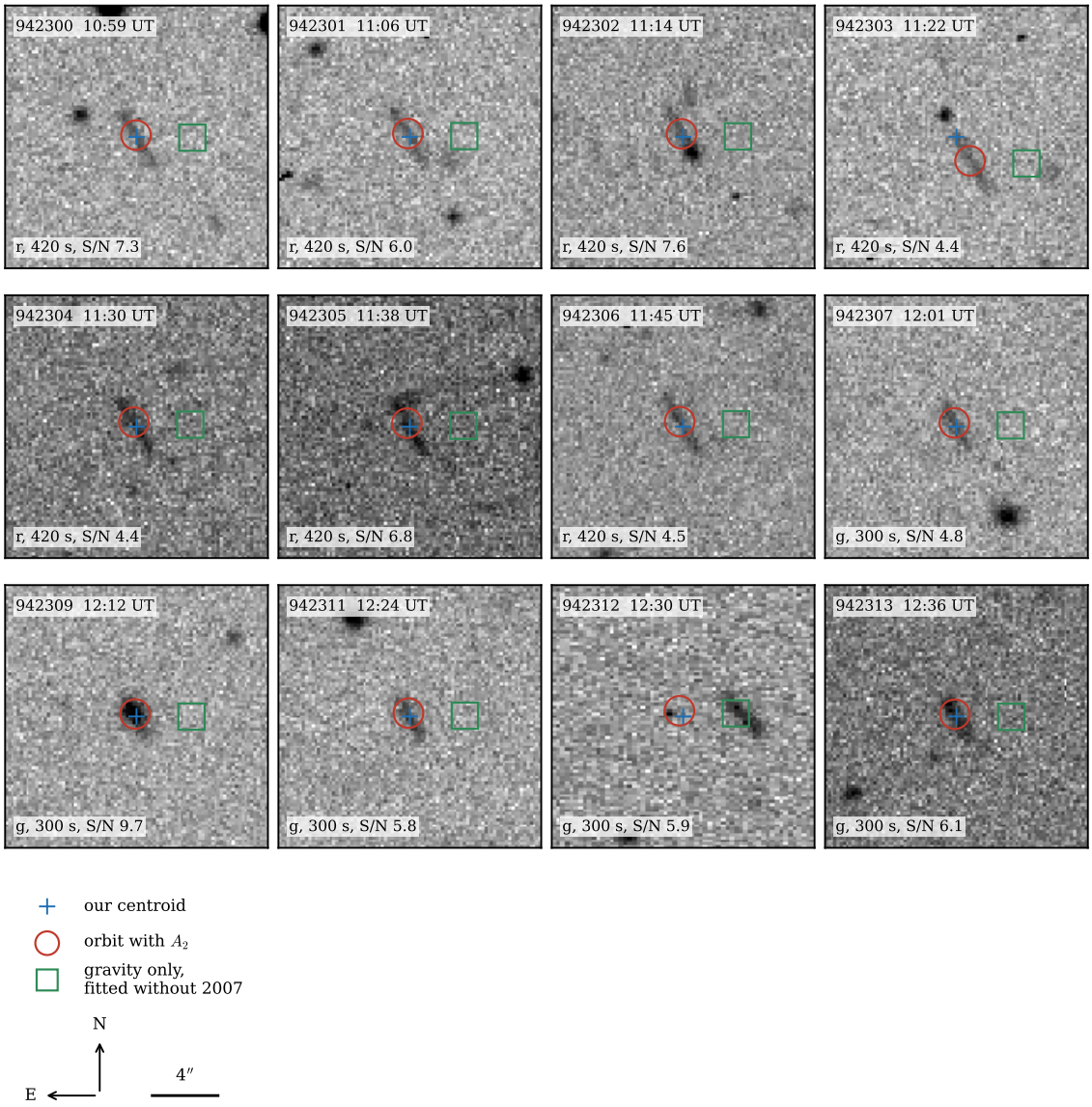


Figure 2: 2016 TA56 in the twelve CFHT MegaCam frames of 2007 September 14 in which it is detected, 16″ across, north up and east left: our centroid (cross), the position predicted by the orbit with A_2 (circle) and the position predicted by the gravity-only orbit fitted to the later apparitions (square). The object is trailed toward the southwest by its motion during each exposure.

Table 3: Our measurements of 2016 TA56 in the CFHT MegaCam frames of 2007 September 14: frame, mid-exposure time (UTC), band, exposure, peak S/N, trail length, position, and its offsets east and north from the orbit with A_2 and from the gravity-only orbit fitted without 2007, seen from site 568.

Frame	UTC	Band	t (s)	S/N	Trail ($''$)	RA (deg)	Dec (deg)	From A_2 orbit ($''$)	From gravity only ($''$)
942300p	10:59:00	r	420	7.3	4.06	36.350460	−4.498394	−0.03, −0.11	+3.40, +0.05
942301p	11:06:48	r	420	6.0	4.06	36.349822	−4.499511	−0.12, −0.21	+3.31, −0.05
942302p	11:14:42	r	420	7.6	4.06	36.349208	−4.500609	−0.10, −0.19	+3.34, −0.03
942303p	11:22:30	r	420	4.4	4.06	36.348850	−4.501241	+0.83, +1.46	+4.26, +1.62
942304p	11:30:18	r	420	4.4	3.94	36.347959	−4.502816	−0.16, −0.28	+3.27, −0.13
942305p	11:38:07	r	420	6.8	4.06	36.347336	−4.503888	−0.18, −0.21	+3.26, −0.06
942306p	11:45:55	r	420	4.5	4.06	36.346710	−4.505005	−0.21, −0.31	+3.22, −0.15
942307p	12:01:19	g	300	4.8	2.90	36.345512	−4.507136	−0.13, −0.23	+3.30, −0.08
942309p	12:12:55	g	300	9.7	2.90	36.344603	−4.508734	−0.09, −0.15	+3.35, +0.01
942311p	12:24:31	g	300	5.8	2.90	36.343682	−4.510371	−0.08, −0.20	+3.35, −0.05
942312p	12:30:19	g	300	5.9	2.81	36.343181	−4.511221	−0.22, −0.35	+3.21, −0.19
942313p	12:36:08	g	300	6.1	2.90	36.342755	−4.511974	−0.10, −0.14	+3.34, +0.02

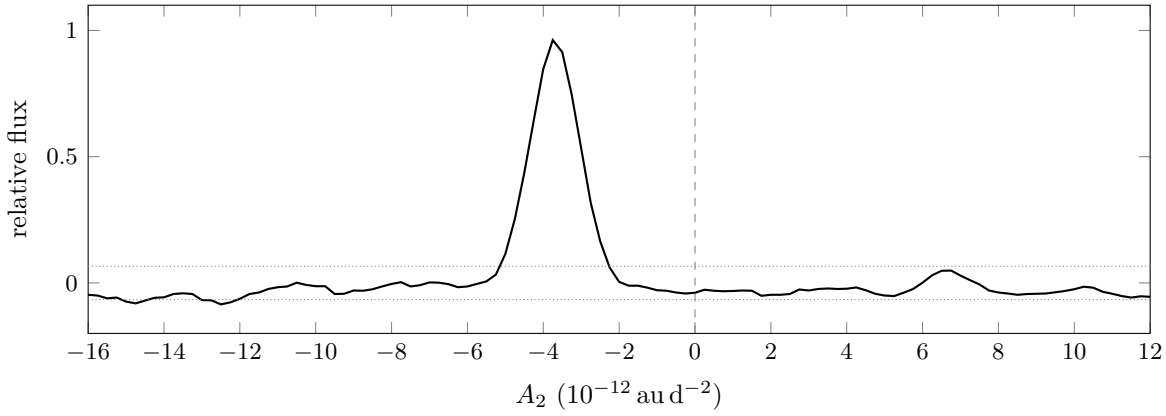


Figure 3: Median flux over the 2007 frames that measure each position, in units of the detection's, of a source moving with 2016 TA56 at each position that orbits fitted to the later apparitions allow, labelled by the value of A_2 that places the object there. The dashed line marks the gravity-only orbit ($A_2 = 0$) and the dotted lines ±3 times the scatter at a fixed point along lines displaced across the field.

Table 4: NEATM fits to the 16 W2 fluxes of 2016 TA56 at fixed beaming parameters, with the efficiency ξ of the refitted A_2 relative to the radiation limit at $1000\,\mathrm{kg\,m^{-3}}$.

η	D (km)	p_V	χ^2	ξ
0.9	1.41	0.050	31.0	14.7
1.0	1.60	0.039	30.2	16.7
1.2	2.00	0.025	29.1	20.8
1.4	2.44	0.017	28.4	25.4
1.6	2.92	0.012	28.0	30.4
1.9	3.72	0.007	27.8	38.7

magnitudes, uncertainties and quality flags, and the use made of each. The JPL database, which includes the NEOWISE diameters from the Planetary Data System, gives none for 2016 TA56; for the dark comet 2005 UY6, which has the same H, it gives 2.25 km and an albedo of 0.018 [12], and for 2012 UR158 0.63 km and 0.023. Albedos near 0.02 are typical of cometary nuclei.

At the fitted diameter the refitted acceleration is $\xi = 12.7$, 25.4 and 50.8 times the radiation limit at 500, 1000 and $2000\,\mathrm{kg\,m^{-3}}$; across the range of η it is 7 to 19 times the limit at $500\,\mathrm{kg\,m^{-3}}$. No bulk density between 500 and $2000\,\mathrm{kg\,m^{-3}}$ brings the acceleration within the radiation limit for any beaming parameter from 0.9 to 1.9.

3.5. Minimum outgassing rate

Gas leaving the surface at about $0.5\,\mathrm{km\,s^{-1}}$ imparts the largest transverse momentum when all of it is emitted along track, so the acceleration sets a lower limit on the production rate, $\dot{M} \geq M A_2\, g(r)/v$. At $D = 2.44\,\mathrm{km}$ and $1000\,\mathrm{kg\,m^{-3}}$ the mass is $7.6 \times 10^{12}\,\mathrm{kg}$ and the limit is $1.1\,\mathrm{kg\,s^{-1}}$ of water ($3.6 \times 10^{25}\,\mathrm{molecules\,s^{-1}}$) at 1 au and $3.1\,\mathrm{kg\,s^{-1}}$ at the perihelion distance of 0.59 au; at $500\,\mathrm{kg\,m^{-3}}$ it is half as large. Gas leaving at an angle θ to the track supplies only $\cos\theta$ of its momentum along it, and the required rate rises by $1/\cos\theta$. These are production rates of weakly active comets. The acceleration sets a lower limit rather than the actual rate, and no coma has been reported for 2016 TA56; the activity of 1998 SH2 was not visible in archival images and was detected only with large-aperture imaging [6].

3.6. The 2030 apparition

The accelerated orbit and the gravity-only orbit fitted without the precovery agree to within $0.2''$ until 2026 August, when the object fades below $V = 22.4$. At the next apparition they diverge: by $6.7''$ on 2030 July 30 at $V = 20.2$ and solar elongation 60°, by $4.0''$ on 2030 September 8 at $V = 21.2$, and by $2.1''$ on 2030 October 18 at $V = 22.4$. Propagating the fit covariances, the predicted position of each orbit is uncertain almost entirely along the direction of separation, by $1.18''$ for the orbit with A_2 and $0.71''$ for the gravity-only orbit on July 30 with one position per night, and by less than $0.1''$ across it (Table 5). A single position accurate to $0.2''$ at any time from July 30 to September 28 therefore distinguishes the two orbits at 4.6 to 4.7σ with one position per night and at 6.9 to 7.5σ with site weights. These significances assume that both orbits are correct up to their fit covariances. The apparition also measures the acceleration without the precovery: added to the 2016 to 2026 positions without the 2007 night, a single position accurate to $0.2''$ on 2030 July 30 gives $\sigma(A_2) = 0.39 \times 10^{-12}\,\mathrm{au\,d^{-2}}$ with one position per night, against 6.8×10^{-12} from those positions alone, and three or six positions spread over the apparition give 0.38 and 0.37×10^{-12}, the uncertainty of the state then dominating. Observations in 2030 can therefore

Table 5: The orbit with A_2 and the gravity-only orbit fitted without 2007 at the 2030 apparition, seen from the geocentre: their separation, the 1σ uncertainty of each along the separation from its fit covariance, and the significance with which a single position accurate to $0.2''$ distinguishes them, with one position per night and with site weights.

	One position a night				Site weights			
Date	Sep. ($''$)	σ_{A_2} ($''$)	$\sigma_{\rm grav}$ ($''$)	σ	Sep. ($''$)	σ_{A_2} ($''$)	$\sigma_{\rm grav}$ ($''$)	σ
2030 July 30	6.55	1.18	0.71	4.7	6.72	0.72	0.49	7.5
2030 August 19	5.36	0.95	0.58	4.7	5.49	0.59	0.40	7.5
2030 September 8	3.89	0.68	0.42	4.7	4.00	0.42	0.29	7.3
2030 September 28	2.76	0.48	0.30	4.6	2.84	0.29	0.20	6.9

confirm or exclude a term of the fitted size at about 9σ whether or not the 2007 source is 2016 TA56. Table S3 of the supplementary material gives daily positions of both orbits from 2030 June to November with their 1σ ellipses. Deep imaging at the same apparition, with the object 1.0 to 1.6 au from the Sun, could detect the coma implied by the acceleration, as large-aperture imaging did for 1998 SH2 [6].

3.7. Overview of the tests

Table 6 lists the main tests of the acceleration of 2016 TA56 and the alternative explanation each addresses. The first six measure the significance of the term and whether measurement errors could produce it, all on the assumption that the 2007 source is 2016 TA56; the significance of the fit does not test that assumption. The search of the allowed positions and the census of moving sources test the identification itself, in the 2007 images. The radiation limit concerns the cause of the measured term, and the 2030 apparition will test the result independently of the precovery.

3.8. 1998 SH2 and 2006 VC

Applying the same fits and thermal model to the two further objects identified by the screen at $2000\,\mathrm{kg\,m^{-3}}$ and albedo 0.14 (Table 7) confirms an acceleration for one of them but not for the other.

(875163) 1998 SH2 ($a = 2.74$ au, $e = 0.71$, $q = 0.79$ au, $i = 2.4°$, $T_J = 2.91$) has 404 positions in three apparitions: 48 at its discovery in 1998 September, 104 from 2016 February to October and 252 from 2025 August to October. With site weights, the fit with A_2 retains 379 positions at an rms of $0.30''$ and gives $A_2 = (-6.11 \pm 0.24) \times 10^{-13}\,\mathrm{au\,d^{-2}}$; with one position per night it gives $(-6.45 \pm 0.54) \times 10^{-13}$, a 12σ detection, and with a radial term (0.4σ) included it gives $(-6.18 \pm 0.30) \times 10^{-13}$, compared with JPL's $(-6.96 \pm 0.49) \times 10^{-13}$. The gravity-only fit retains 354 positions at χ^2/dof 0.90, against 0.67 with A_2. As for 2016 TA56, the earliest apparition deviates most from a gravity-only orbit: the gravity-only orbit fitted to 2016 and 2025, which predicts the 1998 September positions to $0.52''$ along and $0.21''$ across the apparent motion, is offset from the discovery positions by $13.0''$ along and $5.3''$ across the motion, and a timing error cannot produce the cross-track offset. Fits omitting one apparition give $A_2 = -4.3 \pm 1.2$, -1.8 ± 0.6 and $-1.9 \pm 2.1 \times 10^{-12}\,\mathrm{au\,d^{-2}}$, all of the same sign but two of them 2 to 3σ larger than the full-arc value, so the magnitude of the term is less certain than its formal uncertainty. NEOWISE detected the object six times in W2 at 1.14 au from the Sun and phase angles of 58 to 59°: at $\eta = 1.4$ the fit gives $D = 0.64$ km and $p_V = 0.019$, and η from 0.9 to 1.9 gives 0.39 to 0.93 km and 0.051 to 0.009; the NEOWISE catalogue gives 0.38 ± 0.06 km and 0.058 ± 0.024 [12], matching our fit at

Table 6: Tests of the transverse acceleration of 2016 TA56, the alternative each addresses and its result.

Test	Alternative addressed	Result
Fit of all 102 positions	A statistical fluctuation	9σ with each night as one position, 13σ under the night-level covariance, 6.5σ from the night jackknife
Error shared by the 2007 positions	Correlated errors within the precovery night	$1.2''$ needed to bring A_2 below 3σ
2007 offset across the motion	A clock error in the precovery reports	$3.0''$ across the motion, 15 times the prediction uncertainty
Independent measurement of the CFHT frames	An error in either report's reduction or catalogue	The offset is in the images; both reports lie $0.10''$ rms from our track
Radial term fitted with A_2	A radial force mimicking a transverse one	A_1 at 2σ; A_2 unchanged
Gravity-only orbit through 2007 to 2022	A gravity-only orbit that fits the precovery	It misses the 2025 positions by $10.5''$
Positions allowed by the later apparitions	The precovery is another object	One source among them; nothing at the gravity-only position to 3.9 mag below it
Census of moving sources in the 2007 frames	An unrelated object moving as predicted	None within $5''$ per hour and 10° of the motion; 8×10^{-6} expected within $30''$
NEOWISE diameter and the radiation limit	Thermal recoil or radiation pressure	15 to 39 times the limit; significant drifts have a median of 0.075 of it
The 2030 apparition	All of the above	One position gives $\sigma(A_2) = 0.39 \times 10^{-12}\,\mathrm{au\,d^{-2}}$ without the precovery

Table 7: Orbit fits of 1998 SH2 and 2006 VC to the MPC positions, as in Table 2.

Object	Fit	Positions	Kept	χ^2/dof	A_2 (10^{-13} au d^{-2})
1998 SH2	with A_2, site weights	404	379	504.4/751	-6.11 ± 0.24
	with A_2, one position a night	404	401	130.7/795	-6.45 ± 0.54
	with A_1 and A_2	404	379	504.3/750	-6.18 ± 0.30
	gravity only	404	354	634.8/702	—
	without 1998 September	356	342	442.4/677	-42.7 ± 12.3
	without 2016	300	284	328.3/561	-18.0 ± 5.7
	without 2025	152	135	210.4/263	-19.0 ± 21.1
2006 VC	with A_2, site weights	216	185	345.0/363	$+5.54 \pm 0.94$
	with A_2, one position a night	216	213	143.2/419	$+6.66 \pm 1.47$
	gravity only	216	183	343.9/360	—
	without 2025–2026	193	163	316.6/319	$+5.65 \pm 8.65$

$\eta = 0.9$. The acceleration is then 0.7 to 1.7 times the radiation limit at 1000 kg m^{-3} across this range, and 0.7 times at the catalogue diameter. It lies within the limit for the smaller sizes, but only if more than half of the re-emitted momentum were directed along track, whereas the thermal lag of the Yarkovsky effect directs at most about a tenth [21]. A dark body of 0.4 to 0.9 km on a Jupiter-family orbit, with an acceleration that radiation forces cannot plausibly produce, matches the outer dark comets; at the catalogue diameter and 1000 kg m^{-3} the minimum outgassing rate is 8×10^{-4} kg s^{-1} of water at 1 au. Farnocchia et al. [6] showed that it is a comet: they found these perturbations in its 1998–2025 astrometry and, in large-aperture images taken during its 2025 August close approach, a weak tail of low surface brightness not seen in archival images. The screen, with the generic thresholds of Section 2, and the orbit analysis applied to 2016 TA56 therefore recover an independently confirmed, weakly active comet from its acceleration alone; relative to the radiation limit at the same beaming parameter, the acceleration of 2016 TA56 is twenty times larger. The recovery is retrospective, since the activity of 1998 SH2 was reported before the screen was run, so it shows that the screen is sensitive to weakly active comets but is not a blind prediction, and it does not by itself establish that 2016 TA56 is one.

(428209) 2006 VC ($a = 1.94$ au, $e = 0.49$, $q = 0.99$ au, $T_J = 3.72$), not detected by NEOWISE, has 216 positions in four apparitions from 2006 to 2026. The fit with A_2 gives $(+5.5 \pm 0.9) \times 10^{-13}$ au d^{-2}, and $(+6.7 \pm 1.5) \times 10^{-13}$ with one position per night, but the gravity-only fit is equally good (χ^2/dof 0.96 against 0.95, with two fewer positions retained), and omitting the apparition of 2025 October to 2026 May gives $(+5.7 \pm 8.7) \times 10^{-13}$. The term therefore comes from that apparition alone, and the present arc does not require an acceleration. The two orbits diverge from 2033 September, by up to 25″ in 2034 February at $V = 19.8$.

4. Relation to previous work

Farnocchia et al. [5] established the systematic detection of Yarkovsky drifts in near-Earth asteroid astrometry and the comparison of these drifts with the radiation force. Seligman et al. [16] identified the first dark comets among bodies whose accelerations exceed that force, and their later sample [17] separated the inner and outer populations. Taylor et al. [18] traced the dynamical origins of the dark comets and proposed an evolutionary track through rotational fragmentation, and Taylor et al. [19] examined how strong non-gravitational accelerations affect the linking of

observations. Farnocchia et al. [6] attributed the non-gravitational perturbations of 1998 SH2 to cometary activity by imaging its weak tail, suggested that more near-Earth objects classified as asteroids could be comets with undetected outgassing, and called for systematic searches; the screen presented here is such a search over every transverse term in JPL's database. We are not aware of any earlier report of a non-gravitational acceleration of 2016 TA56 beyond JPL's orbit solution.

5. Conclusion

Among the transverse accelerations in JPL's database, that of 2016 TA56 exceeds the radiation limit by a factor that no plausible density or albedo removes, while the significant drifts of the database have a median efficiency of 0.075. The object is on a Jupiter-family orbit like those of the outer dark comets and has not previously been listed among them. Its acceleration is measured conditionally on the 2007 precovery. With that night the fit gives 9σ when each night counts as one position and 13σ under a night-level covariance estimated from the residuals; the uncertainty from refits that each omit one later night corresponds to 6.5σ; and the twelve precovery positions would have to share an error of $1.2''$ to bring the term below 3σ. Without that night the later apparitions do not measure it. The 2007 images identify the precovery source with 2016 TA56: among all positions that orbits fitted to the later apparitions allow, it is the only source, nothing lies at the gravity-only position to 3.9 mag below it, and no other moving source in the frames shares its motion. NEOWISE indicates a dark body about 2.4 km across, similar to the dark comet 2005 UY6. All drifts with known spin poles have the sign expected from the spin. The same screen and orbit analysis recover, retrospectively, the weakly active comet 1998 SH2 from an acceleration 0.7 to 1.7 times the radiation limit at $1000\,\mathrm{kg\,m^{-3}}$; over the same range of beaming parameters the acceleration of 2016 TA56 is 15 to 39 times the limit. The orbit fit measures a transverse term and the thermal fit shows that radiation forces cannot produce it, but no coma has been reported, and outgassing is an inference from the size of the term; 2016 TA56 is a candidate dark comet, not a confirmed comet. The 2030 apparition tests the result independently of the precovery: the two orbits then diverge by 4 to $7''$ at $V \approx 20$ to 21, a single position accurate to $0.2''$ would measure A_2 to $\pm 0.4 \times 10^{-12}\,\mathrm{au\,d^{-2}}$ without the 2007 night, and deep imaging could detect the faint coma that outgassing at this rate implies. For the third object, 2006 VC, the term depends on its latest apparition alone. The screen covers only the 611 asteroids for which JPL has fitted a transverse term, 609 of the 42,772 known near-Earth asteroids among them; it finds candidates among measured accelerations and does not measure the abundance of dark comets.

Acknowledgements

This work uses observations from the Minor Planet Center; orbit solutions and ephemerides from JPL's Solar System Dynamics group through the Small-Body Database and Horizons; the NEOWISE single-exposure source table from the NASA/IPAC Infrared Science Archive; the DAMIT database; and the software ASSIST, REBOUND, astropy, NumPy and SciPy.

Data availability

The analysis software and the derived results are available at

https://github.com/CharlesCNorton/seti-tools

in the directory `papers/dark_comet`, where a README lists each data file behind the tables and figures and the command that produces it. Tables S1 to S3 are in the supplementary material. The observations, orbit solutions, NEOWISE detections, CFHT images and spin models are publicly available from the archives cited in the text.

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